Nationwide TDP Capstone Project 2026 V2.0

Sprint 2 — Carbon Impact & Data Processing

Duration: 3 Days

Objective

Extend the Sprint 1 MVP to calculate and present meaningful **carbon-impact information** from customer spending.

Sprint Goal

A customer can submit spending data and receive a calculated carbon-impact result based on emissions-factor data.

1. Agile & Team Working

- Review and revise the team's **Definition of Ready (DoR)** and **Definition of Done (DoD)**.
- Update the product backlog and sprint backlog.
- Maintain the task board and development log.
- Review team roles, collaboration and responsibilities.

2. Spending & Emissions Data

Extend the database and datasets to support:

- Spending categories.
- Emissions factors.
- Customer spending.
- Data validation and quality checks.
- Appropriate units and calculation rules.

Use SQL/T-SQL to create, query and validate emissions-factor data.

3. Carbon Calculations

Implement deterministic carbon-impact calculations using the approved emissions-factor data.

The application should:

- Receive spending information.
- Identify the relevant spending category.
- Retrieve the appropriate emissions factor.
- Calculate estimated carbon impact.
- Return a clear result to the customer.

The database and controlled calculation logic remain authoritative for carbon calculations.

4. API Development

Extend the API to include:

GET /health

GET /categories

POST /spend

POST /carbon-impact

Validate inputs and handle errors securely.

5. Testing

Extend automated testing to cover:

- Valid spending data.
- Invalid spending data.
- Category validation.
- Emissions-factor retrieval.
- Carbon calculations.
- API responses.
- Database integration.

Include unit and integration testing where appropriate.

6. DevSecOps

Continue the pipeline:



Focus on:

- Secure input validation.
- SQL injection prevention.
- Secure database access.
- Dependency security.
- Secrets management.
- Security-finding remediation.
- Automated testing within CI/CD.

7. Data Learning

Apply relevant learning from:

- **Fundamentals of Data Literacy**
- **Querying SQL Databases using T-SQL**
- **Introduction to Data Science for Data Professionals**
- **Data Handling in Python**
- **Databricks Data Engineering** — stretch

8. Responsible AI Assistance

AI can assist with:

- Code development.
- Test creation.
- SQL queries.
- Data-processing logic.
- Troubleshooting.

- Documentation.

Teams must **review, understand, test and security-check** AI-generated output before using it.

AI should not replace controlled emissions-factor data or authoritative carbon calculations.

9. Sprint 2 Showcase

Demonstrate:

1. Customer spending submission.
2. Input validation.
3. Emissions factor retrieved from the database.
4. Carbon-impact calculation.
5. Carbon-impact result presented to the customer.
6. SQL/T-SQL queries working.
7. Automated tests passing.
8. CI/CD pipeline operational.
9. SAST and SCA completed.
10. DAST completed where practical.
11. Security findings reviewed and addressed.

Sprint 2 Success

The MVP now allows a customer to **submit spending and receive a validated carbon-impact result calculated from controlled emissions-factor data**, supported by automated testing and DevSecOps practices.